FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

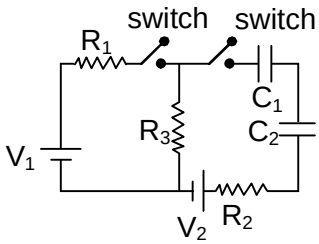
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. In the given circuit $C_1 = C$, $C_2 = 2C$, $V_1 = V_2 = V$ & $R_1 = R_2 = R_3 = R$ and all switches-are closed at t = 0 then choose the incorrect statement at time t = RC (initially all capacitors are uncharged)



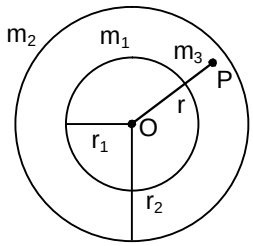
(A) Current through V_1 is less than $\frac{V}{2Re}[e-1]$

(B) Power delivered by V_2 is less than $\frac{V^2}{3eR}$

(C) Current through V_1 is greater than $\frac{V}{2Re}[e-1]$

(D) Power delivered by V_2 is greater than $\frac{V^2}{3eR}$

2. Two concentric spherical shells masses m_1 and m_2 and radii r_1 and r_2. Then



(A) outer shell will have no contribution in gravitational field at point P

(B) force on P is directed towards O

(C) force on P is $\frac{Gm_1m_2}{r^2}$

(D) force on P is $\frac{Gm_1m_3}{r^2}$

3. A ball of mass m is attached to the lower end of light vertical spring of force constant k. The upper end of the spring is fixed. The ball is released form rest with the spring at its normal (un- stretched) length, and comes to rest again after descending through a distance x. Then,

(A) x = mg/k

(B) x = 2mg/k

(C) The ball will have no acceleration at the position where it has descended through x/2.

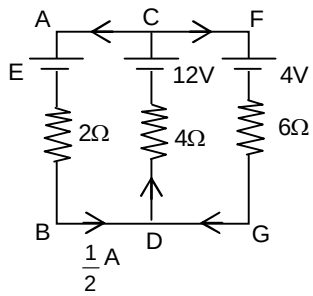
(D) The ball will have an upward acceleration equal to g at its lowermost position

Space for Rough work

4. If the potential difference of Coolidge tube producing X-ray is increased, then choose the correct option(s).
(A) the interval between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ increases
(B) the interval between $\lambda_{k\alpha}$ and λ_0 increases
(C) the interval between $\lambda_{k\beta}$ and λ_0 increases
(D) λ_0 does not change
Here λ_0 is cutoff wavelength and $\lambda_{k\alpha}$ and λ_{kB} are wavelength of k_α and k_B characteristic X-rays.

5. A microammeter has a resistance of 100Ω and a full scale range of 50μA. It can be used as a voltmeter or as a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combinations
(A) 50 V range with 10 kΩ resistance in series
(B) 10 V range with 200 kΩ resistance in series
(C) 5 mA range with 1 Ω resistance in parallel
(D) 10 mA range with 1 Ω resistance in parallel

6. In the circuit shown in figure
(A) E = 6.6 V
(B) i_1 = 1.1 A
(C) i_2 = 0.5 A
(D) E = 4.4 V



7. A closed organ pipe of length 28 cm closed at one end is found to be at resonance when a tuning fork of frequency 850 Hz is sounded near the open end. If velocity of sound in air is 340 m/s, then the
(A) air in the pipe is vibrating in fundamental mode
(B) air in the pipe is vibrating in first overtone
(C) end correction of the pipe is 1 cm
(D) end correction of the pipe is 2 cm

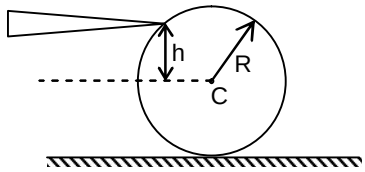
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(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 8, Q. 9 and Q. 10 by appropriately matching the information given in the three columns of the following table.

A billiard ball, initially at rest, is given a sharp impulse by a cue. The cue is held horizontally a distance h above the central line as shown in adjacent figure. The ball leaves the cue with a speed V_0.



Column 1	Column 2	Column 3
(I) Clockwise	(i) $h = \frac{4R}{5}$	(P) Anticlock-wise
(II) Friction acts in forward direction.	(ii) $h = \frac{2R}{5}$	(Q) $t = \frac{2V_0}{7\mu g}$
(III) $\omega = \frac{9V_0}{7R}$	(iii) $\alpha = \frac{5\mu g}{2R}$	(R) Initial angular velocity $\omega_0 = \frac{2V_0}{R}$
(IV) Friction acts in backward direction.	(iv) $\alpha = \frac{2\mu g}{5R}$	(S) Initial angular velocity $\omega_0 = \frac{V_0}{R}$

8. In subsequent forward motion, the ball eventually acquires a final speed of $\frac{9}{7}V_0$, then which one of the combination in true,
(A) (I) (iii) (R) (B) (II) (i) (R)
(C) (III) (i) (S) (D) (IV) (ii) (S)

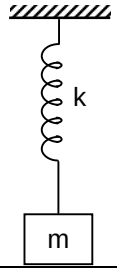
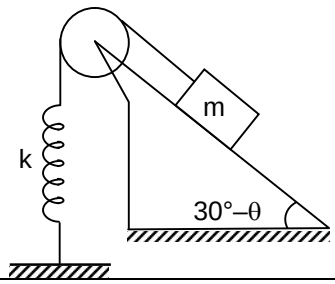
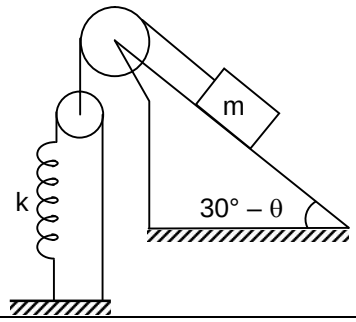
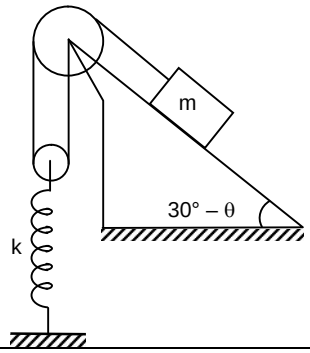
9. If μ is the coefficient of friction between the ball and the surface and α in the angular acceleration of the ball then which combination holds true for μ and α.
(A) (I) (iii) (R) (B) (II) (iii) (P)
(C) (III) (iii) (S) (D) (IV) (iv) (P)

10. If ω is the final angular velocity after time t the ball starts pure rolling, while the coefficient of friction is μ, then which combination holds true.
(A) (III) (iii) (Q) (B) (IV) (i) (Q)
(C) (III) (ii) (R) (D) (III) (ii) (S)

Space for Rough work

Answer questions Q. 11, Q. 12 and Q. 13 by appropriately matching the information given in the three columns of the following table.

Column-1 represents some systems, in which the mass is released for rest with spring relaxed. All contact surface are smooth, strings light, inextensible and pulleys light and smooth. Column-2 contains amplitude of small oscillations and column-3 contains time period for the oscillations.

Column 1	Column 2	Column 3
(I) 	(i) $\frac{mg}{2k}$	(P) $2\pi\sqrt{\frac{2m}{k}}$
(II) 	(ii) $\frac{mg}{k}$	(Q) $\pi\sqrt{\frac{m}{k}}$
(III) 	(iii) $\frac{2mg}{k}$	(R) $4\pi\sqrt{\frac{m}{k}}$
(IV) 	(iv) $\frac{mg}{8k}$	(S) $2\pi\sqrt{\frac{m}{k}}$

Space for Rough work

11. Which of the following options is a correct combination?
(A) (II) (i) (Q) (B) (II) (i) (S)
(C) (III) (ii) (R) (D) (II) (ii) (P)

12. Which of the following options is a correct combination?
(A) (III) (ii) (S) (B) (III) (iv) (R)
(C) (III) (iv) (Q) (D) (III) (ii) (P)

13. Which of the following options is a correct combination?
(A) (IV) (iii) (R) (B) (IV) (iii) (Q)
(C) (IV) (ii) (P) (D) (IV) (i) (S)

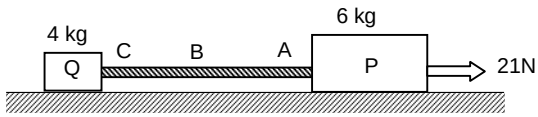
SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. A drop of water of mass m = 0.4 g is placed between two clean glass plates, the distance between which is 0.01 cm. If the force needed to hold one of the plates is 3.2x N, find 'x'. (Surface tension of water = 0.08 N/m).

15. A bus is moving towards a huge wall with a velocity of 5 m/s. the driver sounds a horn of frequency 200 Hz. What is the frequency of beats heard by a passenger of the bus, if the speed of sound in air is 330 m/s?

16. Two blocks of masses 6 kg and 4 kg connected by a rope of mass 4 kg are resting on frictionless floor as shown. If a constant force of 21 Newton is applied to 6 kg block, tension (In N) in the rope at point B is (CB : BA = 1 : 1)



17. Potential energy of a particle moving along x-axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state.

18. A perfectly absorbing surface intercepts a parallel beam of monochromatic light of power 100 W (λ = 500 nm) incident on it normally. If the force exerted by light beam on the surface is $\frac{10^{-6}}{n}$ N, then n =

Space for Rough work

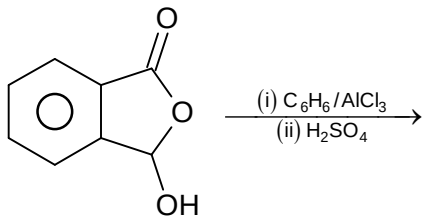
Chemistry

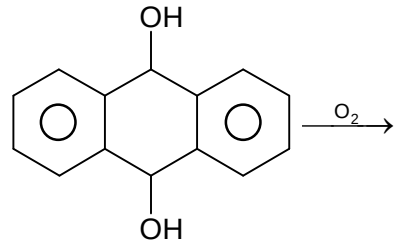
PART – II

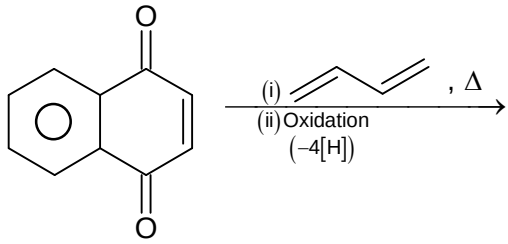
SECTION – A
(One or More than one correct type)

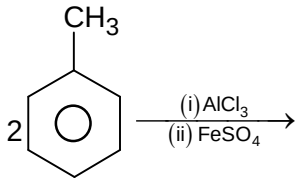
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following reaction(s) will give Anthraquinone as major product?

(A) 

(B) 

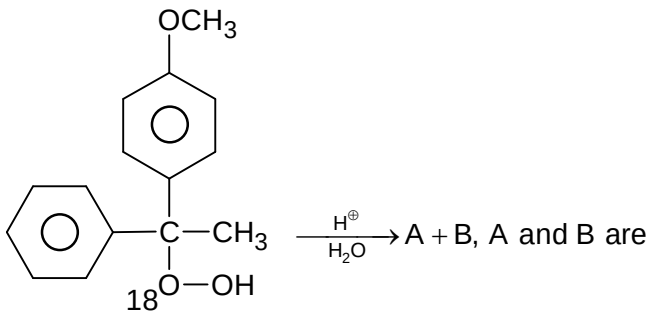
(C) 

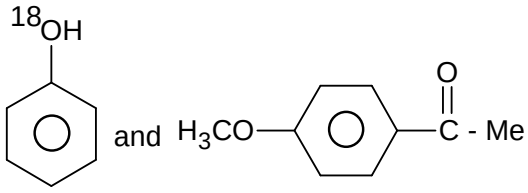
(D) 

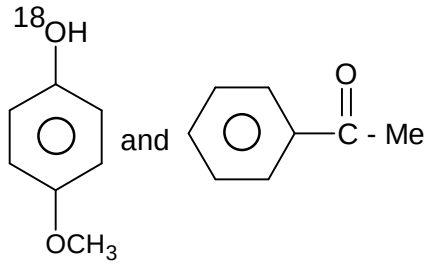
20. When O_2 is adsorbed on a metallic surface, electron transfer occurs form the metal to O_2. The TRUE statement(s) regarding this adsorption is(are)
(A) O_2 is physisorbed
(B) heat is released
(C) occupancy of π^*_{2p} of O_2 is increased
(D) bond length of O_2 is decreased

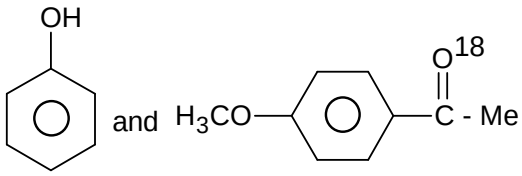
21. Which of the following pairs produce same gas?
(A) (Ca_3N_2 + dil. HCl) and (NH_4NO_2 on heating)
(B) [$(NH_4)_2Cr_2O_7$ on heating] and (NH_4NO_2 on heating)
(C) (NH_4NO_3 on heating) and [$Hg(NO_3)_2$ on heating]
(D) (NH_4Cl on heating) and [$NaNO_3$ + Zn + $NaOH_{(aq.)}$ on heating]

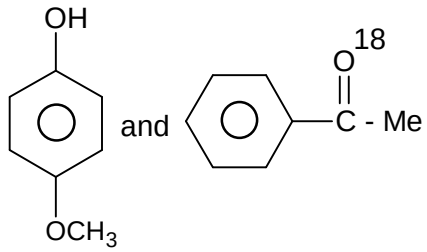
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22. 

(A) 

(B) 

(C) 

(D) 

23. For the reaction:

$I^- + ClO_3^- + H_2SO_4 \longrightarrow Cl^- + HSO_4^- + I_2$

The correct statement(s) in the balanced equation is/are

(A) stoichiometric coefficient of HSO_4^- is 6 (B) iodide is oxidised

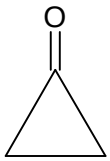
(C) sulphur is reduced (D) H_2O is one of the products

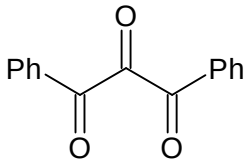
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24. Many carbonyl compounds undergo reversible hydration in aqueous solution as

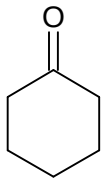
$R_2C=O+H_2O \rightleftharpoons R_2C(OH)_2$

For which of the carbonyl compounds, the above equilibrium is shifted towards the product at 25°C.

(A) 

(B) 

(C) Cl_3CCHO

(D) 

25. Choose the correct relations regarding basic strength?

(A) o-$MeC_6H_4NH_2$ < $C_6H_5NH_2$ < C_6H_5NHMe < p-$MeC_6H_4NH_2$

(B) o-$NO_2C_6H_4NH_2$ < o-$MeC_6H_4NH_2$ < o-$MeOC_6H_4NH_2$ < o-$HOC_6H_4NH_2$

(C) C_4H_5N(Pyrrole) < MeCN < $C_6H_5NH_2$ < C_5H_5N(Pyridine)

(D) All of these

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 26, Q. 27 and Q. 28 by appropriately matching the information given in the three columns of the following table.

Column – 1 (Reaction condition)		**Column – 2 (Intermediate)**		**Column – 3 (Reaction type)**	
(I)	$Me_2C(OH)-CMe_2(NH_2) \xrightarrow[0-5^oC]{NaNO_2+HCl}$	(i)	Carbocation	(P)	Carbyl amine
(II)	$RC(=O)-NH_3 \xrightarrow{\Delta} ? \xrightarrow{H_3O^+} ?$	(ii)	Carbanion	(Q)	Claisen
(III)	$RNH_2 \xrightarrow{CHCl_3/KOH}$	(iii)	Isocyanate	(R)	Pinacol
(IV)	$CH_3COOC_2H_5 \xrightarrow{NaOEt}$	(iv)	Carbene	(S)	Curtius

26. Which of the following is the correct combination?

(A) (I), (i) & (R) (B) (I), (ii) & (R)

(C) (II), (ii) & (S) (D) (II), (i) & (S)

Space for Rough work

27. Which of the following is the correct combination?
(A) (I), (i) & (S) (B) (IV), (ii) & (Q)
(C) (IV), (i) & (Q) (D) (IV), (ii) & (S)

28. Which of the following is the incorrect combination?
(A) (II), (iii) & (S) (B) (III), (ii) & (P)
(C) (III), (iv) & (P) (D) (IV), (ii) & (Q)

Answer questions Q. 29, Q. 30 and Q. 31 by appropriately matching the information given in the three columns of the following table.

Column – 1 (Reaction condition)		Column – 2 (Intermediate)		Column – 3 (Reaction type)	
(I)	White P_4 + alkali	(i)	$H_4P_2O_6$	(P)	+3
(II)	Red P_4 + alkali	(ii)	H_3PO_2	(Q)	+5
(III)	$P_4O_{10} + H_2O$	(iii)	H_3PO_4	(R)	+4
(IV)	$PCl_3 + H_3PO_3$	(iv)	$H_4P_2O_5$	(S)	+1

29. Which of the following is the correct combination?
(A) (I), (ii) & (R) (B) (I), (ii) & (S)
(C) (II), (ii) & (R) (D) (II), (iv) & (R)

30. Which of the following is the correct combination?
(A) (II), (i) & (R) (B) (II), (ii) & (R)
(C) (III), (iii) & (P) (D) (IV), (iv) & (Q)

31. Which of the following is the incorrect combination?
(A) (IV), (iv) & (P) (B) (III), (iii) & (Q)
(C) (II), (i) & (R) (D) (IV), (iv) & (R)

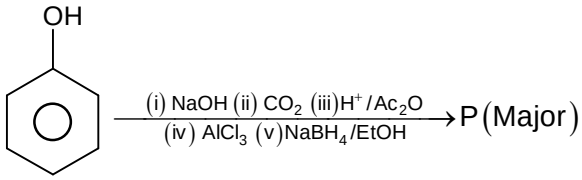
SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0 % to the total weight of the hydrolyzed products. The number of glycine units present in the decapeptide is

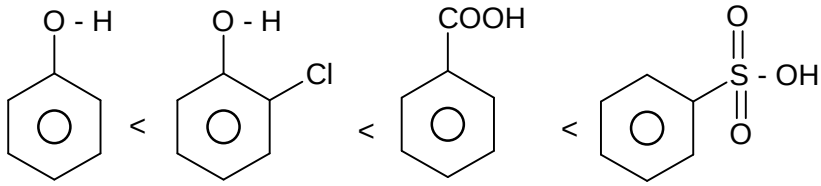
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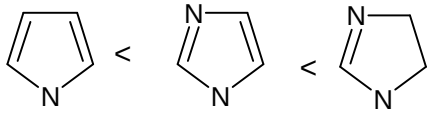
33. No. of S – S linkage in $S_3O_9 = m_1$
No. of P – P linkage in$(HPO_3)_3 = m_2$
No. of S – S linkage in $S_2O_8^{2-} = m_3$
No. of O – O linkage in $S_2O_6^{2-} = m_4$
Find out $m_1 + m_2 + m_3 + m_4$

34. 

Total number of oxygen atoms in the aromatic product(P) is

35 Osmotic pressure of a solution containing non-volatile solute is given by π = CRT. Calculate the work involved (in kcal) when the concentration of a solution at 163 ^oC is changed from 1 M to 0.1 M by adding solvent. [Take R = 2 calorie unit]

36. How many are indicating correct order for given characteristic or property?
(i) $NH_3 > SbH_3 > AsH_3 > BiH_3$ [B.P]
(ii) $CH_4 < CH_3Cl < CH_2Cl_2 < CHCl_3 < CBrCl_3$ [density]
(iii) $CCl_4 < CHCl_3 < CH_2Cl_2 < CH_3Cl$ [dipole moment]
(iv) RF < RCl < RBr < RI [reactivity]
(v) 

[acidic strength]
(vi) 

[basic strength]
(vii) $H_2O < H_2S < H_2Se < H_2Te$ [reducing power]
(viii) $CHF_3 < CHCl_3 < CHBr_3$ [acidic strength]
(ix) RI < RBr < RCl < R – F [thermal stability]

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $\sec(2\theta-2\alpha), \frac{1}{2}\operatorname{cosec}^2\theta, \sec(2\theta+2\alpha)$ are in H.P., then:

(A) $-\frac{\pi}{4}<\alpha<\frac{\pi}{4}$ (B) $\frac{\pi}{4}<\alpha<\frac{3\pi}{4}$

(C) $\frac{3\pi}{4}<\alpha<\frac{5\pi}{4}$ (D) $\frac{5\pi}{4}<\alpha<\frac{7\pi}{4}$

38. Which of the following definite integral(s) vanishes?

(A) $\int_0^{\pi/2} \ln(\cot x)\,dx$ (B) $\int_0^{2\pi} \sin^3 x\,dx$

(C) $\int_{1/e}^{e} \frac{dx}{x(\ln x)^{1/3}}$ (D) $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}\,dx$

39. With usual notation, in a ΔABC the value of $\prod(r_1-r)$ can be simplified as:

(A) $abc\prod\tan\frac{A}{2}$ (B) $4rR^2$

(C) $\frac{(abc)^2}{R(a+b+c)^2}$ (D) $4Rr^2$

40. The value (s) of t for which the lines $2x+3y=5$, $t^2x+ty-6=0$ and $3x-2y-1=0$ are concurrent, can be:

(A) t = 2 (B) t = –3

(C) t = –2 (D) t = 3

Space for Rough work

41. Let ΔPQR be formed by the common tangents to the circles $x^2 + y^2 + 6x = 0$ and $x^2 + y^2 - 2x = 0$, then which of the following is (are) true?

(A) Centroid of ΔPQR is (1, 0)

(B) Radius of circle intscribed in ΔPQR is 1.

(C) Area of ΔPQR is $3\sqrt{3}$

(D) The y – intercept of common tangent having negative slope is 3.

42. If $\left(9+\sqrt{80}\right)^n = I + f$ where I and n are integers and $0 < f < 1$, then:

(A) I is an odd integer

(B) I is an even integer

(C) $(I+f)(1-f)=1$

(D) $1-f=\left(9-\sqrt{80}\right)^n$

43. Let $f(x)=\begin{cases}\dfrac{e^x-1+ax}{x^2}, & x>0\\ b, & x=0\\ \dfrac{\sin\frac{x}{2}}{x}, & x<0\end{cases}$ then:

(A) $f(x)$ is continuous at x = 0 if $a=-1, b=\dfrac{1}{2}$

(B) $f(x)$ is discontinuous at x = 0, if $b\neq\dfrac{1}{2}$

(C) $f(x)$ has non – removable discontinuity at x = 0 if $a\neq -1$

(D) $f(x)$ has removable discontinuity at x = 0 if a = –1, $b\neq\dfrac{1}{2}$.

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 44, Q. 45 and Q. 46 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contains conics, equations of tangents to the conics and points of contact, respectively.

Column 1	Column 2	Column 3
(I) $x^2 + y^2 = 9$	(i) $my = m^2x + 3$	(P) $\left(\frac{3}{m^2}, \frac{6}{m}\right)$
(II) $y^2 = 12x$	(ii) $y = mx \pm \sqrt{9m^2 + 9}$	(Q) $\left(\frac{3m}{\sqrt{1+m^2}}, \frac{-3}{\sqrt{1+m^2}}\right)$
(III) $16x^2 + 9y^2 = 144$	(iii) $y = mx \pm \sqrt{9m^2 + 16}$	(R) $\left(\frac{-9m}{\sqrt{9m^2+16}}, \frac{16}{\sqrt{9m^2+16}}\right)$
(IV) $9x^2 - y^2 = 9$	(iv) $y = mx \pm \sqrt{m^2 - 9}$	(S) $\left(\frac{-m}{\sqrt{m^2-9}}, \frac{-9}{\sqrt{m^2-9}}\right)$

44. Which is correct combination?
(A) (I) (i) (P) (B) (II) (i) (P)
(C) (II) (i) (Q) (D) (III) (ii) (Q)

45. Which is incorrect – combination?
(A) (I) (ii) (Q) (B) (II) (i) (P)
(C) (III) (iii) (R) (D) (IV) (iv) (Q)

46. Which is incorrect combination
(A) (I) (ii) (Q) (B) (III) (iii) (R)
(D) (IV) (iv) (S) (D) (I) (ii) (R)

Space for Rough work

Answer questions Q. 47, Q. 48 and Q. 49 by appropriately matching the information given in the three columns of the following table.

Column 1	Column 2	Column 3
(I) $f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$	(i) $x=\pm 1$	(P) Non Differentiable
(II) $f(x)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$	(ii) $x=0$	(Q) Differentiable
(III) $f(x)=\cos^{-1}\left(2x^2-1\right)$	(iii) $x=2$	(R) Discontinuous
(IV) $f(x)=\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$	(iv) $x=\frac{1}{2}$	(S) Continuous

47. Which is correct combination
(A) (I) (ii) (Q) (B) (I) (i) (P)
(C) (II) (ii) (Q) (D) (II) (ii) (R)

48. Which is correct combination
(A) (II) (ii) (P) (B) (II) (ii) (R)
(C) (II) (ii) (Q) (D) None of these

49. Which is incorrect combination
(A) (I) (ii) (R) (B) (I) (i) (P)
(C) (II) (ii) (S) (D) (IV) (ii) (P)

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. Using the identity, $\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$ or otherwise, if the value of $\sin^4\left(\frac{\pi}{7}\right) + \sin^4\left(\frac{3\pi}{7}\right) + \sin^4\left(\frac{5\pi}{7}\right) = \frac{a}{b}$, where a and b are co – prime then value of a – b is

51. Let $S = \frac{1}{\sin 8^o} + \frac{1}{\sin 16^o} + \frac{1}{\sin 32^o} + ... + \frac{1}{\sin 4096^o} + \frac{1}{\sin 8192^o}$. If $S = \frac{1}{\sin\alpha}$ where $\alpha \in \left(0, 90^o\right)$, then find α (in degree).

52. Let L denotes the number of surjective functions $f : A \to B$, where set A contains 4 elements and set B contains 3 elements.
M denotes number of elements in the range of the function.
$f(x) = \sec^{-1}(\text{sgn}\, x) + \text{cosec}^{-1}(\text{sgn}\, x)$ where $\text{sgn}\, x$ denotes signum function of x.
And N denotes coefficient of t^5 in $\left(1+t^2\right)^5\left(1+t^3\right)^8$. Find the value of $\frac{(LM+N)}{19}$.

53. Let $f(x)$ be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that $\lim\limits_{t\to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$. Find the value of $\lim\limits_{x\to 1} \frac{\ln(f(x)) - \ln 2}{x-1}$.

54. If $\alpha \neq 1$is a fifth root of unity, then find the value of $\log_{\sqrt{3}}\left|1+\alpha+\alpha^2+\alpha^3-\frac{2}{\alpha}\right|$

Space for Rough work